

Solving Two-Step EquationsWorksheet 7A.01.03 **ANSWER KEY**

1. Match each property of equality to the example that shows it. Write the letter of the example on the line.

Every example below starts from $5 = 3 + 2$.

Property of Equality	Letter
Symmetric Property of Equality	B
Addition Property of Equality	D
Subtraction Property of Equality	E
Multiplication Property of Equality	A
Division Property of Equality	C

A.	$5 \times 4 = (3 + 2) \times 4$
B.	$3 + 2 = 5$
C.	$5 \div 4 = (3 + 2) \div 4$
D.	$5 + 4 = 3 + 2 + 4$
E.	$5 - 4 = 3 + 2 - 4$

B is the symmetric property: the two sides trade places, so $5 = 3 + 2$ becomes $3 + 2 = 5$. Nothing is added, removed or scaled. In C the divisor is 4, not 0 — the division property requires a nonzero divisor.

Use the equation $2 + 3n = 11$ for questions 2 through 5.

2. Complete the sentence to describe the equation.

The variable n is first being **multiplied** by 3, and then 2 is being **added** to this product. The value of the expression $2 + 3n$ is **equal** to 11.

3. Describe the operation that isolates $3n$ on the left side.

Subtract 2 from both sides of the equation.

Subtraction property of equality.

4. Describe the operation that isolates n on the left side.

Divide both sides of the equation by 3.

Division property of equality. Multiplying both sides by $1/3$ is equally correct — that is the multiplication property.

5. Now solve $2 + 3n = 11$. Then check your solution by substitution.

$$2 + 3n - 2 = 11 - 2$$

$$3n = 9$$

$$3n \div 3 = 9 \div 3$$

$$n = 3$$

$$\text{Check: } 2 + 3(3) = 2 + 9 = 11 \checkmark$$

Subtraction property, then division property.

For questions 6 through 11, solve the equation and check your solution by substitution.

6. $2x - 5 = 11$

$$2x - 5 + 5 = 11 + 5$$

$$2x = 16$$

$$2x \div 2 = 16 \div 2$$

$$x = 8$$

$$\text{Check: } 2(8) - 5 = 16 - 5 = 11 \checkmark$$

Addition property, then division property.

7. $\frac{n}{12} - 3 = 7$

$$n/12 - 3 + 3 = 7 + 3$$

$$n/12 = 10$$

$$(n/12) \times 12 = 10 \times 12$$

$$n = 120$$

$$\text{Check: } 120 \div 12 - 3 = 10 - 3 = 7 \checkmark$$

Addition property, then multiplication property.

8. $6y + 5 = 29$

$$6y + 5 - 5 = 29 - 5$$

$$6y = 24$$

$$6y \div 6 = 24 \div 6$$

$$y = 4$$

$$\text{Check: } 6(4) + 5 = 24 + 5 = 29 \checkmark$$

Subtraction property, then division property.

9. $9a + 7 = 88$

$$9a + 7 - 7 = 88 - 7$$

$$9a = 81$$

$$9a \div 9 = 81 \div 9$$

$$a = 9$$

$$\text{Check: } 9(9) + 7 = 81 + 7 = 88 \checkmark$$

Subtraction property, then division property.

10. $\frac{y}{4} - 8 = 4$

$$y/4 - 8 + 8 = 4 + 8$$

$$y/4 = 12$$

$$(y/4) \times 4 = 12 \times 4$$

$$y = 48$$

$$\text{Check: } 48 \div 4 - 8 = 12 - 8 = 4 \checkmark$$

Addition property, then multiplication property.

11. $-8 + \frac{b}{3} = 10$

$$-8 + b/3 + 8 = 10 + 8$$

$$b/3 = 18$$

$$(b/3) \times 3 = 18 \times 3$$

$$b = 54$$

$$\text{Check: } -8 + 54 \div 3 = -8 + 18 = 10 \checkmark$$

Addition property, then multiplication property. The variable term is on the right of the constant here, which changes nothing.